Introduction of Course

(Need of Electricity & Magnetism for CS students)

=> Computer involve or comprise of

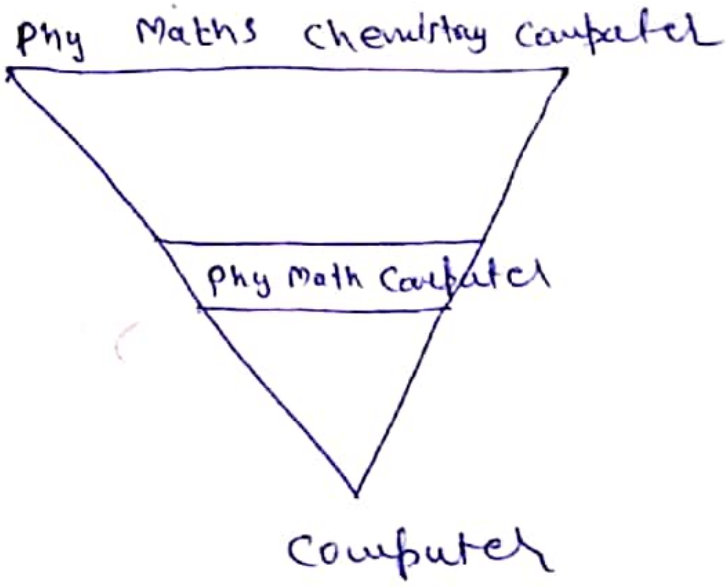
(i) wires (current, Resistance, Heat dissipation)

(ii) Capacitors and inductors

(iii) Material to Extract heat out of the computer, Laptop etc

(iv) charge (electric field)

(v) Current or moving charge (magnetic field)



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Basic Electricity and Magnetism.

Electromagnetism A preview

=> Greeks knew that some stones attract Iron. (600 B.C)

=> In 1820 Hans Cristian Oersted found a relation b/w electricity and magnetism.
"Electric current in a wire can deflect a magnetic compass needle"

=> Micheal Faladay developed electromagnetism and James Clerk Maxwell (1831 - 1879) put faladays idea into mathematical form.

=> Maxwell concluded light is electromagnetic in nature its speed can be deduced from purely electric and magnetic measurements (i.e $v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 m/s$)

=> Heinrich hertz 20 year after Maxwell theory, he produced electromagnetic "Maxwellian waves" now we call radio waves. (Marconic practical application)

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Four kinds of forces (3)

(1) strong forces (Neucleas) (2) Electromagnetic force

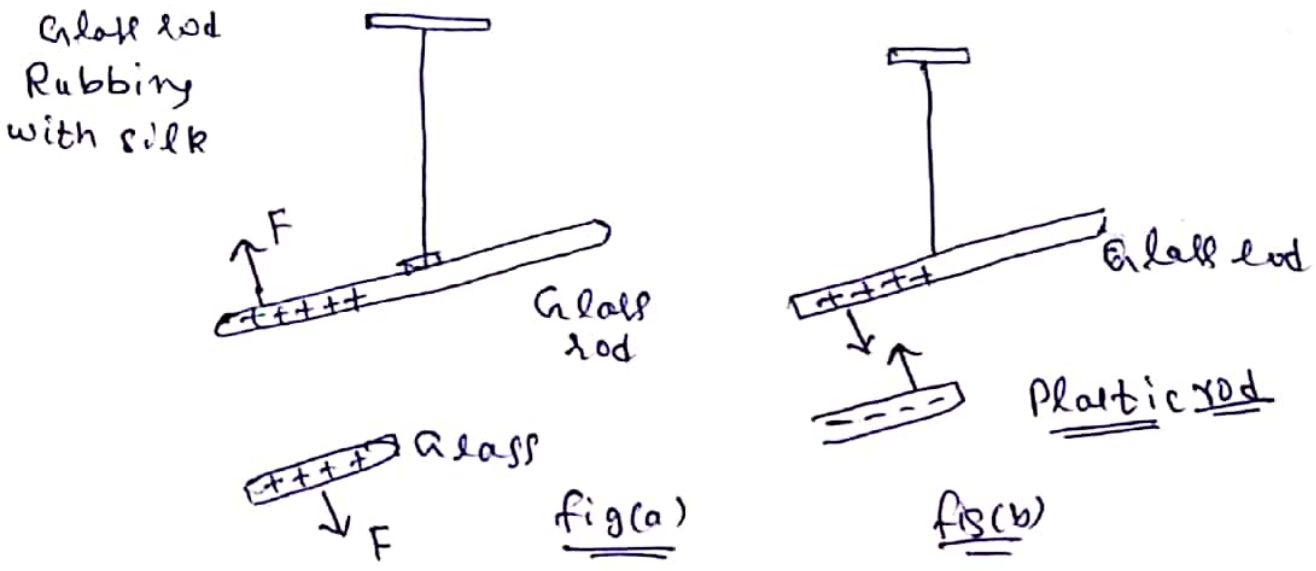
(3) Weak forces (accounts for certain kind of radioactive decay) (4) Gravitational forces.

⇒ Einstein unified field theory was not particularly successful.

⇒ In 1960, Glashow, Weinberg and Salam gave electroweak theory (joins weak and EM forces)

☆ Electric charge

(1) charge of two types ("plus" and "minus")



A body in charge means (imbalance in charge)

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Charges of same sign repel each other and charges of opposite sign attract each other.

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(2) Charge is conserved (it can not be created or destroyed)

$\Rightarrow$ A plus charge can annihilate a minus charge – but plus charge can not simply disappear.

(So total charge is fixed. This is called global conservation)

$\lambda = \frac{dq}{dx}$ $\rho = \frac{dq}{dv}$ $\sigma = \frac{dq}{da}$ $\Rightarrow$ $\frac{\partial \rho}{\partial t} + \nabla \cdot J = 0$ continuity eq. mathematical Statement charge conservation. (Study later)

(3) Charge is Quantized { 1st electric charge was thought to be a continous fluid }

$\Rightarrow$ Electric charge comes only in discreat lumps or integral multiples of the basic unit of charge(e)

If $+e$ for proton

$-e$ for electron

$$\boxed{q = ne} \qquad n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

In which, e is unit of elementy charge

$$e = 1.602 \times 10^{-19} C$$

✓ $+10e$, $-6e$ $\quad$ $\times 1.5e$, $\times 7.3e$

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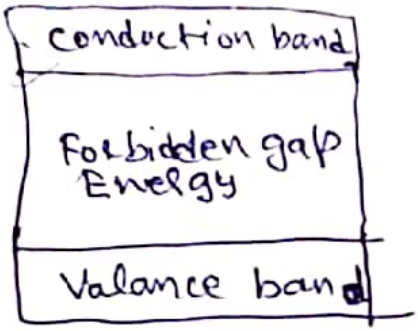
Conductors and Insulators.

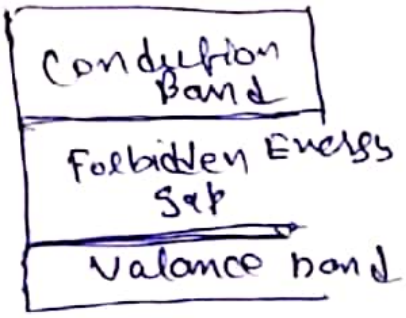
Examples: Copper, Iron Gold and silver good conductor. (charge flow easily)

Insulators

Examples: Glass, chemically pure water and plastic are insulators.

=> Each atom may contribute one conduction electron (in typical conductor)







- Insulator

no. conduction electrons in an insulator

- Semiconductor

$10^{10} - 10^{12}$ conduction electrons per cm^3 Silicon or Germanium

- Conductors

10^{23} Conduction electrons per cm^3

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Coulomb's Law

$F \propto q_1 q_2$ — (1) {SI unit of charge in C}

$F \propto \frac{1}{r^2}$ — (2)

$F \propto \frac{q_1 q_2}{r^2}$ — (3)

$F = K\frac{q_1 q_2}{r^2}$ — (4) $\left(K = \frac{1}{4\pi\varepsilon_0}\right)$

$K = \frac{1}{4\pi\varepsilon_0} = 8.89 \times 10^9 \ N \cdot m^2/C^2$

$\varepsilon_0 = 8.85 \times 10^{-12} \ C^2/N m^2$

⇒ Difference b/w gravitation and electrostatic force is that gravitational force is always attractive while electrostatic force may be repulsive or attractive depend on the charges.

{ New gravitation Law $F = G\frac{m_1 m_2}{r^2}$ }

Coulomb's law : vector Form :-



Charge same sign (Repulsive)



fig (a)

Charge of opposite (Attractive)

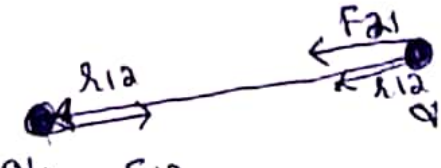


fig (b)

origin of coordinate system at location of particle 2

r_{12} = position vector of particle (1)

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} \quad \text{---------- (5)}$$

F_{12} = Force on particle one(1) from particle 2

$$\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}} \quad \text{---- (6)}$$

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21} \quad \text{---- (7)}$$

$\vec{F}_{21}$ = Force on particle 2 from particle 1

$\Rightarrow$ The vector form of coulomb's law is useful because it contain direction information about F and whether the force is attractive or repulsive.

$$F_1 = F_{12} + F_{13} + F_{14} + \text{------} \quad \text{---- (8)}$$

F_{13} = Force on particle 1 from particle 3

Eq. (8) is mathematical representation of the principle of superposition applied to electric forces.

It permits us to calculate the force due to any of charges if other charges were not present

principle of superposition does not hold in many situations, particularly in strong field.

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